

THE USE OF INFORMATION SETS IN DECODING CYCLIC CODES

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A class of decoding algorithms using encoding-and-comparison is considered for error-correcting code spaces. Code words, each of which agrees on some information set for the code with the word r to be decoded, are constructed and compared with r . An operationally simple algorithm of this type is studied for cyclic code spaces A . Let A have length n , dimension k over some finite field, and minimal Hamming distance m . The construction of fewer than $n^2/2$ code words is required in decoding a word r . The procedure seems to be most efficient for small minimal distance m , but somewhat paradoxically it is suggested on operational grounds that it may prove most useful in those cases where m is relatively large with respect to the code length n .

1. Introduction

Let $R = R(q, n)$ be the (free) concrete n -dimensional vector space over the finite field $GF(q)$ of q elements. Let $A = A(q, n, k)$ be a k -dimensional subspace of R . Let R have the Hamming weight and distance functions $w(r)$ and $d(r, s)$. Let m be the minimal distance between distinct elements of A , and let $t = \lfloor (m-1)/2 \rfloor$ be the packing radius for A .

In decoding a received element (word) r in R , we wish to solve the equation

$$r = b + e, \quad (1.1)$$

where b is a word in the code space A closest to the word r and, equivalently, e is a word of minimal weight in the coset $A+r$. Then r is mapped onto (decoded as) b .

A decoding algorithm is an operational recipe for accomplishing this in whole or part. One classification of such algorithms is by the size of the subset S of R such that each word s in S is correctly mapped onto a closest word in A . For example, the subset S of R might be any of the following: (1.2) all of R ; (1.3) all words r in R such that there is a unique word in the code A closest to r ; (1.4) all words r in R such that there is a word in A (necessarily unique) within distance t of r .

Another classification of decoding algorithms serves to place the present paper. Most decoding algorithms give methods of solving (1.1) by determining the error vector e . Slepian(5) places this problem in a suitable general framework. Outstanding examples of the class are the general

decoding algorithms for the Bose-Chaudhuri codes given in Peterson(3, Sec. 9.4-5).

Here we take a more simple-minded approach. Given a word r to be decoded, we construct a set of words in A that have some chance of being close to r since each constructed word agrees with r on some information set for A . The word r is mapped onto a closest word b among the constructed words if the distance $d(r, b)$ is small enough. In the next section we consider the general ideas related to such a process.

2. Information Sets

We number the coordinate indices in the space R from 0 to $n-1$. By an information set α for A , we mean any subset of the set of coordinate indices such that there is a unique word b in A with any given set of coordinate values b_i , b_i in $GF(q)$, i in α . Clearly there are exactly $k = \dim A$ distinct coordinate indices in any information set α for A . By L_α we shall mean the linear transformation of R onto A such that rL_α is the unique element b of A satisfying $r_i = b_i$, all i in α . A basic property of information sets is the following:

(2.1). Let r be any word in R . Then as α runs through the information sets for A , the set $\{rL_\alpha\}$ contains all elements b in A such that the distance $d(r, b)$ is less than the minimal distance m in A .

Proof. We use the following well known property: If β is any subset of the coordinate indices for R that contains at least $n-(m-1)$ distinct indices, then β includes an information set α for A . Suppose now that $r = b + e$, for b in A and $w(e)$ less than m . Then the set β of coordinate indices i such that $e_i = 0$ contains at least $n-(m-1)$ distinct indices. Thus β includes an information set α for A . Then $rL_\alpha = bL_\alpha + eL_\alpha = b + 0 = b$, proving (2.1).

It is easy enough to find other characterizations of information sets that are equivalent to the definition we have given, but little seems to have been done even in counting the number of distinct information sets for a code space of some specific kind. The number of distinct information sets for a code can be very large. Theorem (2.1) says that an algorithm using all information sets for a code A would work correctly, although it might well be operationally prohibitive. In the next section we state a

decoding algorithm for cyclic codes that makes explicit use only of the information set containing the first k coordinate indices.

3. The Affine Group on J_n

(The reader is referred to Sections 5 and 6. The first summarizes some properties of cyclic codes. The second considers some operational aspects of the algorithm studied. These sections are to be regarded as appendices.)

We now assume that $R(q, n)$ is the algebra over $GF(q)$ whose ideals are cyclic codes. We make the further assumption that n is an odd prime that does not divide q . It is possible to relax the latter requirement, but the facts are both simpler and more favorable if we make it.

We identify the coordinate indices $0, 1, \dots, n-1$ with the elements of the field J_n of integers modulo n . Any permutation π of the coordinate indices induces a coordinate permutation P of R :

$$rP = s \quad \text{such that } r_i = s_{i\pi}. \quad (3.0)$$

In particular let S and T be the coordinate permutations of R induced respectively by the permutations $i \rightarrow i+1$ and $i \rightarrow gi$ of J_n , where g is some generator of the multiplicative group of non-zero elements of J_n .

The group (S) generated by the shift permutation S is of order n . If P is in (S) , then P maps every cyclic code A in R onto itself.

The group (T) generated by the permutation T is of order $n-1$. If P is in (T) , then P is an automorphism of R as an algebra. In particular, P maps every cyclic code A in R upon an equivalent cyclic code, possibly itself.

The group (S, T) generated by the permutations S and T is a metacyclic group of order $n(n-1)$. It is isomorphic to the affine group of permutations $i \rightarrow ai + b$, $a \neq 0$, on J_n . We order the permutations P in (S, T) by

$$P_{ni+j} = T^i S^j \quad \text{for } 0 \leq i \leq n-2, 0 \leq j \leq n-1 \quad (3.1)$$

Thus P runs through the group (S, T) in the order $I, S, S^2, \dots, S^{n-1}, T, TS, \dots, TS^{n-1}, T^2, T^2S, \dots$

If $A(q, n, k)$ is a cyclic code in R , we write L_A for the linear transformation of R onto A such that if $rL_A = b$, then $r_i = b_i$ for $0 \leq i \leq k-1$. Note that any k successive coordinate indices form an information set for a cyclic code.

We can now state a decoding algorithm for the class of cyclic codes we are considering:

(3.2). To decode the word r in R . Let $[z]$, $[d]$, and $[b]$ designate memory registers with initial values 0, n , and 0 respectively.

- (i). For $P = P_z$, compute the distance $\tilde{d} = d(rP, rPL_AP)$. If $\tilde{d} \leq t$, where $t = \lfloor (n-1)/2 \rfloor$, map r onto $\tilde{b} = rPL_AP^{-1}$ and stop. Otherwise go to (ii).
- (ii). If $\tilde{d} < d$, store $\tilde{d}$ in $[d]$, $\tilde{b}$ in $[b]$, and go to (iii). Otherwise go to (iii).
- (iii). If $z+1 = n(n-1)/2$, go to (iv). Otherwise store $z+1$ in $[z]$ and go to (i).
- (iv). If d is less than a threshold τ , map r onto b and stop. Otherwise, stop without decoding r .

This flow analysis omits some operational simplifications that are not conceptually significant. In particular, a remark in Sec. 6 explains why $n^2/4$ rather than $n^2/2$ can be taken as an upper bound on the number of encodings required in decoding r . More important is the question of efficiency: on how large a subset of R does the procedure decode correctly?

4. Gap Length

By the gap length $\gamma(r)$ of a word r in R is meant the largest number of consecutive coordinates of r equal to zero. We write $\delta(r) = n - \gamma(r)$ for the complement of the gap length.

The permutation group (S, T) divides R into transitivity sets (equivalence classes) such that the words r and s are in the same class if and only if $rP = s$ for some P in (S, T) . We write $[r]$ for the equivalence class containing r . Let H be the subset of R containing all elements of weight h , that is, the set of words in R that have exactly $n-h$ coordinates equal to zero. Then we define

$$\begin{aligned} \gamma([r]) &= \text{maximum } \gamma(s) \text{ for } s \text{ in } [r], \\ \delta([r]) &= \text{minimum } \delta(s) \text{ for } s \text{ in } [r], \\ \gamma(n, h) &= \text{minimum } \gamma([r]) \text{ for } r \text{ in } H, \\ \delta(n, h) &= \text{maximum } \delta([r]) \text{ for } r \text{ in } H. \end{aligned} \quad (4.1)$$

Note that in any of the three senses defined γ and δ are complementary, $\gamma + \delta = n$. Note also that these concepts depend only on the distinction between zeros and nonzeros among coordinates of words in R . Thus they are independent of the particular ground field $GF(q)$ over which R is defined.

Before investigating gap length, we show its relevance to the efficiency of the decoding procedure (3.2). Suppose that $r = b + e$, where b is in A . Then by a computation similar to that in the proof of (2.1), we see that b is in the set $\{rPL_AP^{-1}\}$ as P runs through the group (S, T) if and only if $k \leq \gamma([e])$, that is, if and only if there is some permutation P in (S, T) such that the first k coordinates of eP are equal to zero. We also have

(4.2). If $k \leq Y(n, h)$ [equivalently, if $\delta(n, h) \leq n - k$], then the set $\{rPL_{APL}^{-1}\}$ contains all words b in A whose distance from r is less than or equal to h .

If A is a cyclic code with parameters (q, n, k, m, t) , we would like the inequality

$$k \leq Y(n, t) \quad (4.3)$$

to be satisfied. There are two problems involved in investigating (4.3).

Problem 1. We do not have an effective procedure for calculating the minimal distance m for cyclic codes; that is, for calculating the packing radius $t = \lfloor (m-1)/2 \rfloor$. The outstanding result so far is the Bose-Chaudhuri lower bound on minimal distance. Further progress is being made on this central problem in the theory of cyclic codes.

Problem 2. We do not have an effective procedure for computing $Y(n, h)$. In stating some known properties, it will be easier to consider $\delta(n, h) = n - Y(n, h)$.

Note first that any shift permutation in (S) has no effect on Y or δ . Moreover, only half the permutations in (T) can be considered effective. For (T) contains the reversing permutation Q corresponding to the permutation $i \rightarrow -i$ of J_n . If $P' = PQ$, then P and P' have the same effect on gap length. This explains step (iii) of the algorithm, which stops on reaching the permutation Q , half way through the group (S, T) .

(4.4). The inequality $h \leq \delta(n, h)$ holds, with equality in case h is 0, 1, 2, $n-2$, $n-1$ or n .

Proof. The second part of the statement follows from the fact that the affine group on J_n is doubly transitive. A consequence of the fact that $\delta(n, 2)$ equals 2 is that the procedure (3.2) always works for double error correction.

A better lower bound on $\delta(n, h)$ can be obtained by a simple counting argument:

$$(4.5). \text{ Let } \delta = \delta(n, h). \text{ Then } \binom{\delta-1}{h-1} \geq \binom{n}{h} / \binom{n}{2}.$$

Proof. Let $\tilde{H}$ be the set of the $\binom{n}{h}$ words in R with h coordinates equal to 1, $n-h$ coordinates equal to 0. Let $\bar{H}$ be the subset of $\tilde{H}$ containing all words r with $r_0 = 1$, $r_i = 0$ for $i \geq \delta(n, h)$. There are $\binom{\delta-1}{h-1}$ words in $\bar{H}$. The group (S, T) divides $\tilde{H}$ into transitivity sets $[r]$. Each of these transitivity sets contains at least one word in $\bar{H}$ by the definition of δ . The number of words in a transitivity set divides the order $n(n-1)$ of the group (S, T) . If the order of $[r]$ is equal to $n(n-1)$, then $[r]$ contains at least two words in $\bar{H}$; for in this case any word s in $[r]$ is distinct from every shift of its reverse sQ . Thus (4.5) holds.

We give some tables of values of $\delta(n, h)$ that have been computed.

TABLE 4.6. $\delta(23, h)$

h	0	1	2	3	4	5	6	7	8	9	10	11
δ	0	1	2	6	10	12	15	15	17	17	19	19
h	12	13	14	15	16	17	18	19	20	21	22	23
δ	19	19	20	20	20	21	21	21	21	21	22	23

TABLE 4.7. $\delta(n, h)$

h	3	4	5	6	7	8
n						
23	6	10	12	15	15	17
29	6	10	14	17		
31	7	12	16	18	21	22
37	8	13	17	20		
41	8	14	21	23		
43	8	14	20	23		
47	8	15	21	25		
59	9	18	24	30		
71	10	20	28			
73	10	20	28			
89	11	24	33			
113	12	26				
127	14	28	43			
149	14	34				
173	14	35				
199	16	39				

TABLE 4.8. Specific Codes

q	n	k	m	t	$Y(n, t)$
2	23	12	7	3	17
2	31	16	7	3	24
2	41	22	9	4	27
2	47	24	11	5	26
2	127	92	11	5	84

Some values of $\delta(n, h)$ that look out of line when plotted are nevertheless correct. The value $\delta(n, h)$ marks the tail of the distribution of the values $\delta([r])$ for the transitivity sets $[r]$ in H .

Table 4.8 applies these values in considering some specific cyclic codes. Note that in every case except the last the inequality $Y(n, t) \geq k$ is satisfied with room to spare. In these cases the decoding procedure works correctly well beyond the packing radius t .

For the last code listed in Table 4.8, we have $k = 92$, $Y(n, t) = 84$, $Y(n, t-1) = 99$. Indeed only 87% (13,894 out of 15,903) of the transitivity sets containing words of weight t have gap length greater than or equal to k .

I would suggest that the algorithm given may prove most useful when n and m are fairly large. In this case a code somewhat better than required by the channel might be used with somewhat inefficient decoding. This must be considered conjectural.

We note that $\delta(n, m)$ is greater than $n-k$. For if b is any nonzero element of A , and P is in (S, T) , then bP is in a code equivalent to A and must have degree at least $n-k$. We want $\delta(n, t) \leq n-k$. The example for $n = 127$ shows that this cannot be true in general.

5. Cyclic Codes

A mapping $PL_{AP}P^{-1}$ of (3.2) is of course a mapping of the type L_a of (2.1), where a is some information set for the cyclic code A . However this decomposition of L_a as a product requires us to consider mappings $L_{AP} = L_\beta$, where β is the information set containing the first k coordinate indices for a cyclic code AP equivalent to A . One aim of the sketch of cyclic codes given in this section is to throw light on the relations involved here. The reader is referred to Peterson(3, Ch. 8-9) and to Prange(4). Our approach differs technically from that of Peterson in its emphasis on the role of multiplicative identity e_A of a cyclic code A , and in its avoidance of the concept of a splitting field for the polynomial X^n-1 over $GF(q)$.

We start with the set R of n -tuples $r = (r_0, \dots, r_{n-1})$ over the finite field $GF(q)$, considered as a vector space over $GF(q)$. Let $x^{(i)}$ for $0 \leq i \leq n-1$ be the element of R such that $x_i^{(i)} = 1$, the remaining coordinates being equal to zero. Defining a multiplication $x^{(i)} x^{(j)} = x^{(i+j)}$ on these basis elements, where $i+j$ is taken modulo n , makes R an algebra. The element $x^{(0)} = \bar{1} = (1, 0, \dots, 0)$ is the multiplicative identity of the algebra. If we write x for $x^{(1)}$, any element r in R can be represented as a polynomial $r(x) = r_0 + r_1 x + \dots + r_{n-1} x^{n-1}$ in R , and the multiplication in the algebra is polynomial multiplication modulo the relation $x^n = \bar{1}$.

We identify the coordinate indices in R with the ring J_n of integers modulo n . If we consider the coordinate permutation S of R induced by the permutation $i \rightarrow i+1$ of J_n , we see that $rS = x r(x)$. Thus the subspaces of R that are closed under S are exactly the ideals A of R as an algebra. We call the ideals cyclic codes.

We write J_n^* for the multiplicative group generated by the elements of J_n that correspond to integers relatively prime to n . Let P be a permutation of R induced by a permutation $i \rightarrow ai$, where a is in J_n^* . Then $rP = r(x^a)$, and we see that P is an automorphism of R as an algebra. In particular, P maps every ideal in R upon an ideal.

We now assume that q and n are relatively prime. Then q may be regarded as an element of J_n^* , and we see that the permutation P of R induced by $i \rightarrow qi$ is an automorphism of every ideal of R , since $r(x^q) = (r(x))^q$. More generally this holds true for all elements in the subgroup of J_n^* generated by q . We call this subgroup $J_{n,q}^*$.

In this case there are two elements in any ideal A of R that are of particular interest. We call an element of R monic if its first coordinate r_0 is equal to one. Then every ideal A contains a unique monic element f_A of least degree. Such an element corresponds to a factor of X^n-1 over $GF(q)$. If k is the dimension of A as a vector space, then the degree of f_A is $n-k$.

If n is relatively prime to q , any ideal A of R has a multiplicative identity e_A (necessarily unique). Many computations concerning A are most easily related to e_A .

Where A is an ideal in R , we define the complementary ideal A' as the set of all r such that $rb = 0$ for all elements b in A . If e_A is the multiplicative identity of A , then $\bar{1}-e_A$ is the multiplicative identity of A' . The elements f_A and $f_{A'}$ correspond to complementary factors of X^n-1 over $GF(q)$.

Since any ideal A is also a subspace of R , we can consider the orthogonal subspace $A^\perp$ of A containing all elements r such that $b \cdot r = 0$ for all b in A , where $b \cdot r$ is the usual inner product $b_0r_0 + \dots + b_{n-1}r_{n-1}$. Then $A^\perp$ is also an ideal in R , and A' and $A^\perp$ are related by $A'Q = A^\perp$, $A^\perp Q = A'$, where Q is the reversing permutation of R induced by the permutation $i \rightarrow -i$ of the coordinate indices.

The computational importance of the multiplicative identities of the ideals in R is in part due to the following observation: If Z is any automorphism of R as an algebra, then the ideal A is mapped by Z onto the ideal whose multiplicative identity is $(e_A)Z$.

To get a concrete handle on the elements e_A of R , we go to the subalgebra C of R containing all elements r such that $r = r^q$. An element r is in C if and only if r is invariant under the coordinate permutation of R induced by the permutation $i \rightarrow qi$ of the coordinate indices. The subalgebra C has two sets of bases as a vector space over $GF(q)$ that are of particular interest: the set of multiplicative identities for the minimal ideals of R ; the set of elements whose coordinates r_i are equal to one if i is in some transitivity set of J_n under the permutation $i \rightarrow qi$, and equal to zero elsewhere.

The second basis is concrete enough. To make the first basis more concrete, we use the pairing $A \rightarrow A'$ of an ideal with its complementary ideal. In particular, if A is a minimal ideal, then A' is a maximal ideal. The monic element f_A of least degree in a maximal ideal corresponds to an irreducible factor of $X^n - 1$ over $GF(q)$.

Once we know how to change from one basis of C to the other, many computational problems become simple. We give a concrete example.

Example 5.1. Let $n = 11$ and $q = 3$. The elements of R corresponding to monic irreducible factors of $X^n - 1$ over $GF(q)$ are $f_1 = (1, 2, 0, \dots, 0)$, $f_2 = (1, 1, 2, 1, 0, 2, 0, \dots, 0)$ and $f_3 = (1, 0, 2, 1, 2, 2, 0, \dots, 0)$. Let A_i be the minimal ideal such that f_i is in the maximal ideal A_i' . Let e_i be the multiplicative identity of A_i .

The transitivity sets of J_n under the permutation $i \rightarrow qi$ are the sets $\tau_1 = \{0\}$, $\tau_2 = \{1, 3, 9, 5, 4\}$, $\tau_3 = \{10, 8, 2, 6, 7\}$. Let t_i be the element of C whose j th coordinate is equal to 1 if j is in τ_i , zero otherwise.

Change of basis in C from the e_i to the t_i is given by the linear transformation M , where the matrices of M and M^{-1} are $\begin{bmatrix} 222 \\ 110 \\ 101 \end{bmatrix}$ and $\begin{bmatrix} 111 \\ 202 \\ 220 \end{bmatrix}$.

Let A be the ideal $A_1 \cup A_2$, and let e be the multiplicative identity of A . Then e has the expression $(1, 1, 0)$ over the basis e_i and the expression $(0, 0, 2) = (1, 1, 0)M$ over the basis t_i . Thus $e = (0, 0, 2, 0, 0, 0, 2, 2, 2, 0, 2)$. We can take 2 as a generator of J_n^* . Let T be the coordinate permutation of R corresponding to $i \rightarrow 2i$. Then eT has the expression $(0, 2, 0)$ over the basis t_i . We see that $AT = A_1 \cup A_3$, $AT^2 = A$.

We mention two promising areas for research in cyclic codes.

Area 1. The coordinate permutations of R corresponding to the permutations $i \rightarrow ai$ of J_n for a in J_n^* set up equivalence classes of cyclic codes. To study the difference in error-correcting properties for nonequivalent cyclic codes of the same dimension k .

Area 2. Let $G(A)$ be the subgroup of J_n^* such that for all a in $G(A)$ the coordinate permutation corresponding to the permutation $i \rightarrow ai$ of J_n is an automorphism of A . Then $G(A)$ always includes $J_{n,q}^*$ if q is relatively prime to n . Study the codes for which $G(A)$ is a particular subgroup of J_n^* . As an example, suppose that n is prime and that $G(A)$ is the group of squares in J_n^* . If any such codes exist, there are exactly four, two equivalent codes of dimension $(n-1)/2$ and two of

dimension $(n+1)/2$. They are not Bose-Chaudhuri codes in general. The known properties of these codes, which include the Golay code with $(q, n, k) = (2, 23, 12)$, are of great interest. The first four codes of Table 4.8 fall in this class.

6. Notes on the Algorithm

(6.1). The basic operation of the algorithm is encoding-and-comparison. The work in doing one such operation is essentially a linear function of $n-k$. See Peterson(3, Sec. 8.4-5) and Meggitt (2, P. 235) for operational discussions of encoding in cyclic codes.

(6.2). An upper bound on the number of encodings required is $n(n-1)/2$, half the order of the group (S, T) . The following observation allows us to multiply this by $(q-1)/q$, which is $1/2$ for $q = 2$. Suppose we have compared the pair r and rL_β where β is the information set containing the first k coordinate indices. In rS , the last coordinate of r has been shifted into the first position. The last coordinate of r equals the last coordinate of rL_β with probability $1/q$. If this should be true, the comparison of rS with rSL_β can be omitted.

(6.3). The permutation T is defined by some choice of a generator g of the group J_n^* , n an odd prime. Extensive tables of g exist. See Albert(1, p. 157).

(6.4). The number of codes AP equivalent to A as P ranges through (S, T) is the smallest positive integer c such that g^c is in $G(A)$. As P runs through (T) , these codes are entered in regular order.

If -1 is not a square in J_n^* , we can replace the group (T) by the group generated by T^2 without loss, running through the entire group (T^2) . If q is a square in J_n^* , this change cuts the number c of codes equivalent to A entered in the process by a factor of 2.

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